1 Multiple Choice 1 point

Which of the following best describes Artificial General Intelligence (AGI)?

- ○ A calculator that performs complex math
- ● An AI system that can generalize learning across different tasks like a human
- ○ A chatbot that responds to FAQs
- ○ A self-driving car

2 Multiple Choice 1 point

Which of these AI tools is best associated with image recognition?

- ○ Traditional logic programming
- ○ Symbolic reasoning
- ● Deep learning neural networks
- ○ Heuristic search

3 Multiple Choice 1 point

Which of the following is NOT a component of an intelligent agent?

- ○ Sensors
- ○ Actuators
- ● Emotions
- ○ Decision-making mechanism

4 Multiple Choice 1 point

Who is credited with the concept of the Turing Test?

- [x] Alan Turing
- [ ] Marvin Minsky
- [ ] John McCarthy
- [ ] Geoffrey Hinton

5 Multiple Choice 1 point

Which best describes the relationship between AI, ML, and DL?

- [x] DL ⊂ ML ⊂ AI
- [ ] AI ⊂ ML ⊂ DL
- [ ] ML ⊂ DL ⊂ AI
- [ ] ML = AI = DL

6 Multiple Choice 1 point

Which is a real-world example of ML in healthcare?

- [ ] Writing prescriptions
- [x] Using training data to detect cancer in X-rays
- [ ] Sending appointment reminders
- [ ] Calculating patient bills

7 Multiple Choice 1 point

Which ethical concern is most related to biased training data in AI?

- ○ Overfitting
- ○ Lack of transparency
- ◉ Fairness
- ○ Speed of execution

8 Multiple Choice 1 point

In which year was the term “Artificial Intelligence” first coined?

- ○ 1950
- ◉ 1956
- ○ 1965
- ○ 1980

9 True or False 1 point

AI today can match human general intelligence across all domains.

- ○ True
- ◉ False

10 True or False 1 point

A self-driving car using computer vision is an example of Deep Learning.

- (●) True
- ( ) False

11 True or False 1 point

All AI systems are designed with ethical frameworks.

- ( ) True
- (●) False

12 True or False 1 point

Neural networks are designed to simulate the human brain’s structure.

- (●) True
- ( ) False

13 Essay 2 points Short Answer

In your own words, define Machine Learning. Provide one example of how it is used in a STEM field.

14 Essay 2 points Short Answer

Describe what an intelligent agent is. Give an example from a real-world system.

15 Essay 2 points Short Answer

What is one ethical challenge that arises from using AI in hiring or admissions?

16 Essay 2 points Short Answer

List two differences between Machine Learning and Deep Learning.

17 Essay 5 points Scenario-Based Question - Long Answer

Scenario: A self-driving car must choose between hitting a pedestrian or swerving into a wall, potentially harming the driver.
Question: Discuss this from an AI ethics perspective. What principles should guide this decision, and who should be accountable?

18 Essay 5 points Scenario-Based Question - Long Answer

Scenario: A school uses AI to automatically assign grades based on student behavior and performance data.
Question: What are some ethical concerns with this approach?

19 Essay 5 points Scenario-Based Question - Long Answer

Scenario: A facial recognition system in a public space is used for law enforcement.
Question: What are two potential benefits and two ethical risks of this use of AI?

20 Essay 5 points Scenario-Based Question - Long Answer

Scenario: A company uses an AI chatbot to replace human customer service agents.
Question: What are the potential social and economic impacts of this decision?